The documentation and process conversion measures necessary to comply with this revision shall be completed by 27 September 2007.

INCH-POUND

MIL-PRF-19500/413E
27 June 2007
SUPERSEDING
MIL-PRF-19500/413D
21 March 2006

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-POWER,
TYPES 2N3771 AND 2N3772, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon, high-power transistors for use in high-speed power-switching applications. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (TO-3).

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type	P_T		$R_{\theta JC}$	$R_{\theta JA}$	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_{STG} and T_J
	T_A = +25°C (1)	T_C = +25°C (2)								
	W	W	°C/W	°C/W	V dc	V dc	V dc	V dc	A dc	°C
2N3771	6	150	1.17	29.2	50	40	7	7.5	30	-65 to +200
2N3772	6	150	1.17	29.2	100	60	7	5.0	20	-65 to +200

(1) Derate linearly 34.2 mW/°C for T_A > +25°C.
(2) Derate linearly 857 mW/°C for T_C > +25°C.

1.4 Primary electrical characteristics.

Type	h_{FE2} at V_{CE} = 4 V dc				$V_{CE(SAT)1}$ (1)	
	I_C = 15 A dc		I_C = 10 A dc		I_C = 15 A dc, I_B = 1.5 A dc	I_C = 10 A dc, I_B = 1.0 A dc
	Min	Max	Min	Max	V dc	V dc
2N3771	15	60			1.5 Max	
2N3772			15	60		1.2 Max

(1) Pulsed (see 4.5.1).

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

AMSC N/A

FSC 5961

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.

1.4 Primary electrical characteristics - continued.

Limits	$\|h_{fe}\|$	C_{obo}	Switching (see figure 2)	
	V_{CE} = 4 V dc I_C = 1 A dc f = 100 kHz	V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	t_{on}, t_{off} 2N3771	t_{on}, t_{off} 2N3772
			µs µs	µs µs
Min	6			
Max	30	1,200 pF	10 12	8 10

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

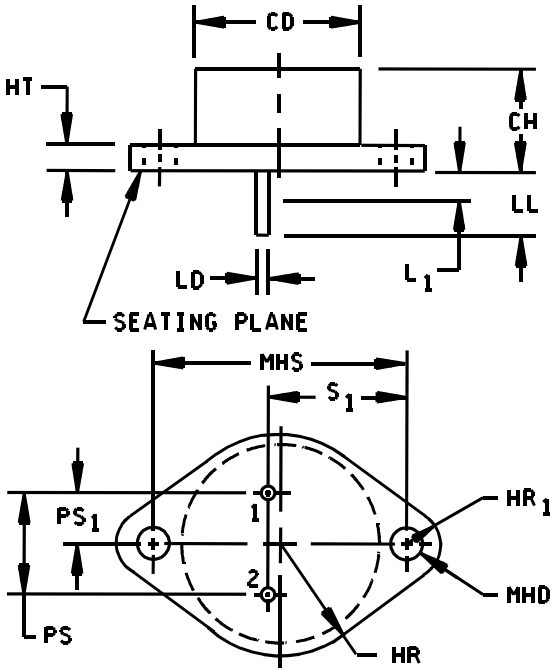
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.

Ltr	Dimension				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CH	.270	.380	6.86	9.65	
CD		.875		22.22	
HR	.495	.525	12.57	13.33	
HR_1	.131	.188	3.33	4.78	
HT	.060	.135	1.52	3.43	
LD	.038	.053	0.97	1.35	7
LL	.312	.500	7.92	12.70	7
L_1		.050		1.27	7
MHD	.151	.165	3.84	4.19	
MHS	1.177	1.197	29.90	30.40	
PS	.420	.440	10.67	11.18	4
PS_1	.205	.225	5.21	5.72	4
S_1	.655	.675	16.64	17.15	



NOTES:

1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Terminal 1 is base; terminal 2 is emitter; case is collector.
4. These dimensions should be measured at points .050 - .055 inch (1.27 mm - 1.40 mm) below seating plane. When gauge is not used, measurement will be made at seating plane.
5. The seating place of the header shall be flat within .004 inch (0.102 mm) inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .004 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
6. Collector shall be electrically connected to the case.
7. LD applies between L_1 and LL. Lead diameter shall not exceed twice LD within L_1.
8. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions, TO-3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, the country of origin marking may be omitted from the body of the transistor.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.

* 4.3 Screening. Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurement JANTX and JANTXV levels
(1) 3c	Thermal impedance (see 4.3.2)
9	I_{CEX1}.
11	I_{CEX1} and h_{FE2}; ΔI_{CEX1} = 100 percent of initial value or 200 μA dc, whichever is greater.
12	Burn-in (see 4.3.1).
13	Subgroup 2 of table I herein. ΔI_{CEX1} = 100 percent of initial value or 500 μA dc, whichever is greater; Δh_{FE2} = ±25 percent of initial reading.

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: $T_A \leq +35°C$; V_{CB} = 25 V dc ±5 V dc; T_J = +187.5°C ±12.5°C. NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW}, (and V_H where appropriate). Measurement delay time (t_{MD}) = 70 μs max. See table III, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1. Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein. End-point electrical measurements shall be in accordance with table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta requirement shall be in accordance with table II herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.

* 4.4.2.1 Group B inspection, table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1037	For solder die attach: Intermittent operation life; 2,000 cycles; $V_{CB} \geq 10$ V dc; $T_A \leq +35°C$
B3	1027	For eutectic die attach: $T_A \leq +35°C$, $V_{CB} \geq 10$ V dc; adjust P_T to achieve $T_J = +175°C$ minimum.
B3	2037	Bond strength; test condition A or C as applicable. All internal leads for each device shall be pulled separately.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta requirements shall be in accordance with table II herein.

Subgroup	Method	Condition
C2	2036	Test condition A, weight = 10 lbs., t = 15 s.
C5	3131	See 4.3.2, $R_{\theta JC} = 1.17°C/W$..
C6	1037	For solder die attach: Intermittent operation life; 6,000 cycles; $V_{CB} \geq 10$ V dc; $T_A \leq +35°C$
C6	1027	For eutectic die attach: $T_A \leq +35°C$, $V_{CB} \geq 10$ V dc; adjust P_T to achieve $T_J = +175°C$ minimum.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta requirements shall be in accordance with table II herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.

* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 2/	3131	See 4.3.2	$Z_{\theta JX}$			°C/W
Breakdown voltage, collector to base	3011	Bias condition D, pulsed (see 4.5.1), I_C = 200 mA dc	$V_{(BR)CEO}$			
2N3771				40		V dc
2N3772				60		V dc
Breakdown voltage collector to emitter	3011	Bias condition B, I_C = 200 mA dc, R_{BE} = 100Ω, pulsed (see 4.5.1)	$V_{(BR)CER}$			
2N3771				45		V dc
2N3772				70		V dc
Breakdown voltage, collector to emitter	3011	Bias condition A, I_C = 200 mA dc, V_{BE} = -1.5 V dc, pulsed (see 4.5.1)	$V_{(BR)CEX}$			
2N3771				50		V dc
2N3772				90		V dc
Collector-emitter cutoff current	3041	Bias condition D	I_{CEO}			
2N3771		V_{CE} = 30 V dc			5	mA dc
2N3772		V_{CE} = 50 V dc			5	mA dc
Emitter to base cutoff current	3061	Bias condition D, V_{BE} = 7.0 V dc	I_{EBO}		2.0	mA dc
Collector-emitter cutoff current	3041	Bias condition A, V_{BE} = 1.5 V dc	I_{CEX1}			
2N3771		V_{CE} = 50 V dc			500	µA dc
2N3772		V_{CE} = 100 V dc			500	µA dc
Base emitter voltage (nonsaturated)	3066	Test condition B, V_{CE} = 4 V dc, pulsed (see 4.5.1),	V_{BE}			
2N3771		I_C = 15 A dc			2.3	V dc
2N3772		I_C = 10 A dc			2.0	V dc
Collector to emitter voltage (saturated)	3071	Pulsed (see 4.5.1)	$V_{CE(sat)1}$			
2N3771		I_C = 15 A dc, I_B = 1.5 A dc			1.5	V dc
2N3772		I_C = 10 A dc, I_B = 1.0 A dc			1.2	V dc

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.

* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector to emitter voltage (saturated)	3071	Pulsed (see 4.5.1)	$V_{CE(sat)2}$			
2N3771		I_C = 30 A dc, I_B = 6 A dc			4	V dc
2N3772		I_C = 20 A dc, I_B = 4 A dc			4	V dc
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), V_{CE} = 4 V dc, I_C = 1.0 A dc	h_{FE1}	40		
Forward current transfer ratio	3076	Pulsed (see 4.5.1), V_{CE} = 4 V dc	h_{FE2}			
2N3771		I_C = 15 A dc		15	60	
2N3772		I_C = 10 A dc		15	60	
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = -1.5 V dc	I_{CEX2}			
2N3771		V_{CE} = 50 V dc			1.5	mA dc
2N3772		V_{CE} = 100 V dc			1.5	mA dc
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), V_{CE} = 4 V dc	h_{FE3}			
2N3771		I_C = 15 A dc		10		
2N3772		I_C = 10 A dc		10		
Subgroup 4						
Pulse response	3251	Test condition A except test circuit and pulse requirements in accordance with figure 2 herein				
Turn on time		V_{CC} = 30 V dc	t_{on}			
2N3771		I_C = 15 A dc, I_{B1} = 1.5 A dc			10	µs
2N3772		I_C = 10 A dc, I_{B1} = 1 A dc			8	µs
Turn off time		V_{CC} = 30 V dc	t_{off}			
2N3771		I_C = 15 A dc, I_{B1} = 1.5 A dc I_{B2} = -1.5 A dc			12	µs
2N3772		I_C = 10 A dc, I_{B1} = 1 A dc I_{B2} = -1 A dc			10	µs

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.

* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 4 V dc, I_C = 1.0 A dc f = 100 kHz	$\|h_{fe}\|$	6	30	
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc, I_C = 1.0 A dc f = 1 kHz	h_{fe}	40		
Output capacitance (open circuit)	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		1,200	pF
Subgroup 5						
Safe operating area (dc operation)	3051	T_C = +25°C, t = 1 s, 1 cycle, see figure 3				
Test 1 (2N3771 only)		I_C = 30 A dc V_{CE} = 5 V dc				
Test 2 (2N3771 only)		I_C = 3.75 A dc V_{CE} = 40 V dc				
Test 3 (2N3772 only)		I_C = 20 A dc V_{CE} = 7.5 V dc				
Test 4 (2N3772 only)		I_C = 2.5 A dc V_{CE} = 60 V dc				
Safe operating area (clamped inductive)	3053	Load condition B, T_C = +25°C, duty cycle ≤ 10 percent, (vary pulse width to achieve I_C), R_s = 0.1Ω, (see figure 4)				
Test 1 (2N3771 only)		R_{BB1} = 2Ω, V_{BB1} ≤ 14 V dc, R_{BB2} = 100Ω, V_{BB2} ≤ 1.5 V dc, I_C = 30 A dc, V_{CC} = 20 ±5 V dc, R_L ≤ .67Ω, L = 5 mH, 0.01 (Signal Transformer Co. CH-30 or equivalent), CR = 1N1186A, clamp voltage = 50 +0, -5 V dc, (device fails if clamp voltage not reached)				

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.

* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Test 2 (2N3772 only)		R_{BB1} = 2Ω, V_{BB1} ≤ 10 V dc, R_{BB2} = 100Ω, V_{BB2} = 1.5 V dc, I_C = 20 A dc, V_{CC} = 40 ±5 V dc, R_L ≤ 2Ω, L = 5 mH, 0.01Ω (Signal Transformer Co. CH-30 or equivalent), CR = 1N1186A, clamp voltage = 90 +0, -5 V dc, (device fails if clamp voltage not reached)				
Electrical measurements		See table I, subgroup 2				
Subgroup 6						
Safe operating area (unclamped inductive)	3053	Load condition C (unclamped inductive load) see figure 5 herein; T_C = +25°C; duty cycle ≤ 10%; R_S = 0.1Ω; R_{BB2} = 100Ω; V_{CC} ≤ 15 V dc.				
Test 1 (2N3771 only)		V_{BB2} = 1.5 V dc; R_{BB1} = 1Ω, V_{BB1} ≤ 12 V dc, I_C = 30 A dc, L = 1 mH, 0.005Ω (Signal transformer Co. CH-100 or equivalent), t_p ≈ 5 ms.				
Test 2 (2N3772 only)		V_{BB2} = 1.5 V dc, R_{BB1} = 2Ω, V_{BB1} ≤ 12 V dc, I_C = 20 A dc, L = 2 mH, 0.01Ω (Signal Transformer Co. CH-50 or equivalent), t_p ≈ 5 ms.				
* Test 3 (both types)		V_{BB2} = 0 V, R_{BB1} ≤ 30Ω, V_{BB1} ≤ 10 V dc, I_C = 5.5 A dc, L = 40 mH, 0.3Ω (Signal Transformer Co. CH-8 or equivalent), t_p ≈ 20 ms.				

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.

* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 6 - Continued						
Safe operating area (clamped inductive) (destructive)	3053	Load condition B, T_C = +25°C, duty cycle ≤ 10%, R_S = 0.1Ω, (see figure 4).				
Test 1 (2N3771 only)		R_{BB1} = 2Ω, V_{BB1} ≤ 14 V dc, R_{BB2} = 100Ω, V_{BB2} = 1.5 V dc, I_C = 30 A dc, V_{CC} = 50 +0,-5 V dc, R_L = 1.67Ω, L = 5 mH, 0.01Ω (Signal Transformer Co. CH-30 or equivalent), CR = 1N1186A, Clamp voltage = 50 +0, -5 V dc.				
Test 2 (2N3772 only)		R_{BB1} = 2Ω, V_{BB1} ≤ 10 V dc, R_{BB2} = 100Ω, V_{BB2} = 1.5 V dc I_C = 20 A dc, V_{CC} = 90 +0,-5 V dc, R_L = 4.5Ω, L = 5 mH, 0.01Ω (Signal Transformer Co. CH-30 or equivalent), CR = 1N1186A, Clamp voltage = 90 +0, -5 V dc.				
Electrical measurements		See table I, subgroup 2				

1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

TABLE II. Groups B, C, and E delta measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1.	Forward-current transfer ratio 2N3771 2N3772	3076	V_{CE} = 4 V dc, pulsed (see 4.5.1) I_C = 15 A dc I_C = 10 A dc	Δh_{FE2} 4/			±25% change from previously measured value

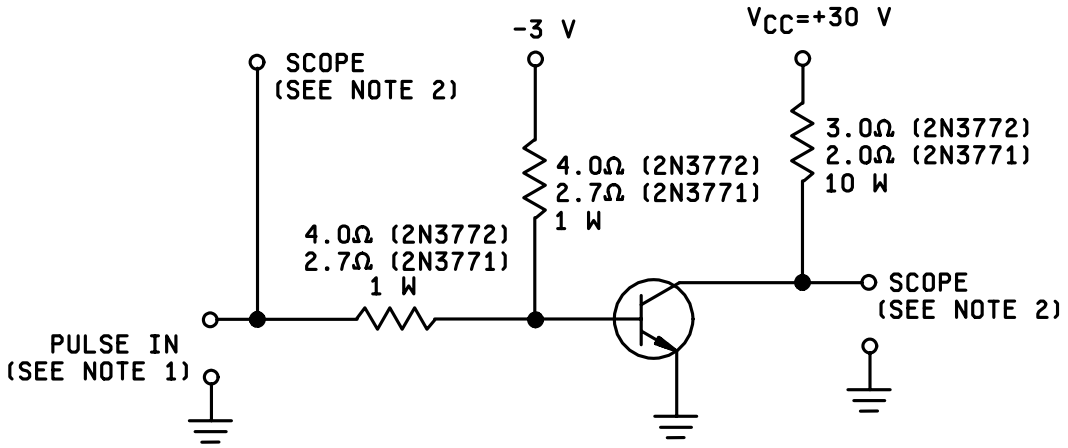
1/ The delta measurements for table E-IX of MIL-PRF-19500 are subgroups 1 and 2.
2/ The delta measurements for table E-VIB, (JAN, JANTX and JANTXV) of MIL-PRF-19500 are subgroups 3 and 6.
3/ The delta measurements for table E-VII of MIL-PRF-19500 are subgroups 2 and 6.
4/ Devices which exceed the group A limits for this test shall not be acceptable.

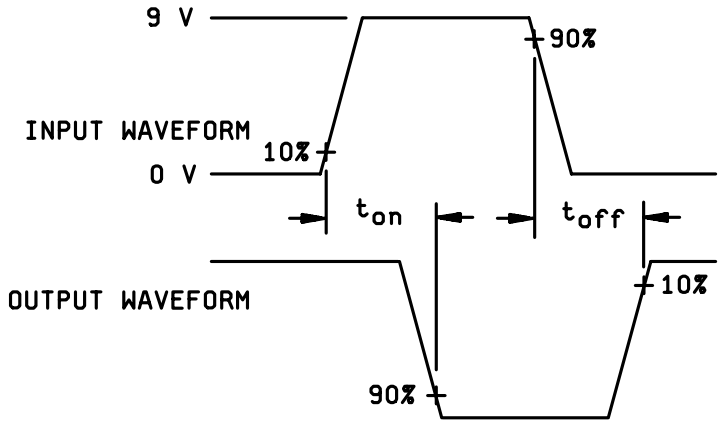
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.

TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	1,000 cycles	
Hermetic seal Fine leak Gross leak	1071	 Test conditions G or H Test conditions C or D	
Electrical measurements		Table I, subgroup 2 and table II herein.	
Subgroup 2			45 devices c = 0
Blocking life	1048	1,000 hours minimum, T_A = +150°C, V_{CB} = 80 percent of rated.	
Electrical measurements		Table I, subgroup 2 and table II herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 6			3 devices
ESD	1020	Testing is not required for class 3 listing. Testing is required for a nonsensitive listing to prove capability.	
Subgroup 8			22 devices c = 0
Reverse stability	1033	Condition B	

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.





NOTES:

1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be each ≤ 20 nanoseconds, duty cycle ≤ 2 percent, generator source impedance shall be 50Ω, pulse width = 20 μs.
2. Output sampling oscilloscope: Z_{in} ≥ 100 kΩ, C_{in} ≤ 50 pF, rise time ≤ 2.0 nanoseconds.

FIGURE 2. Pulse response test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.

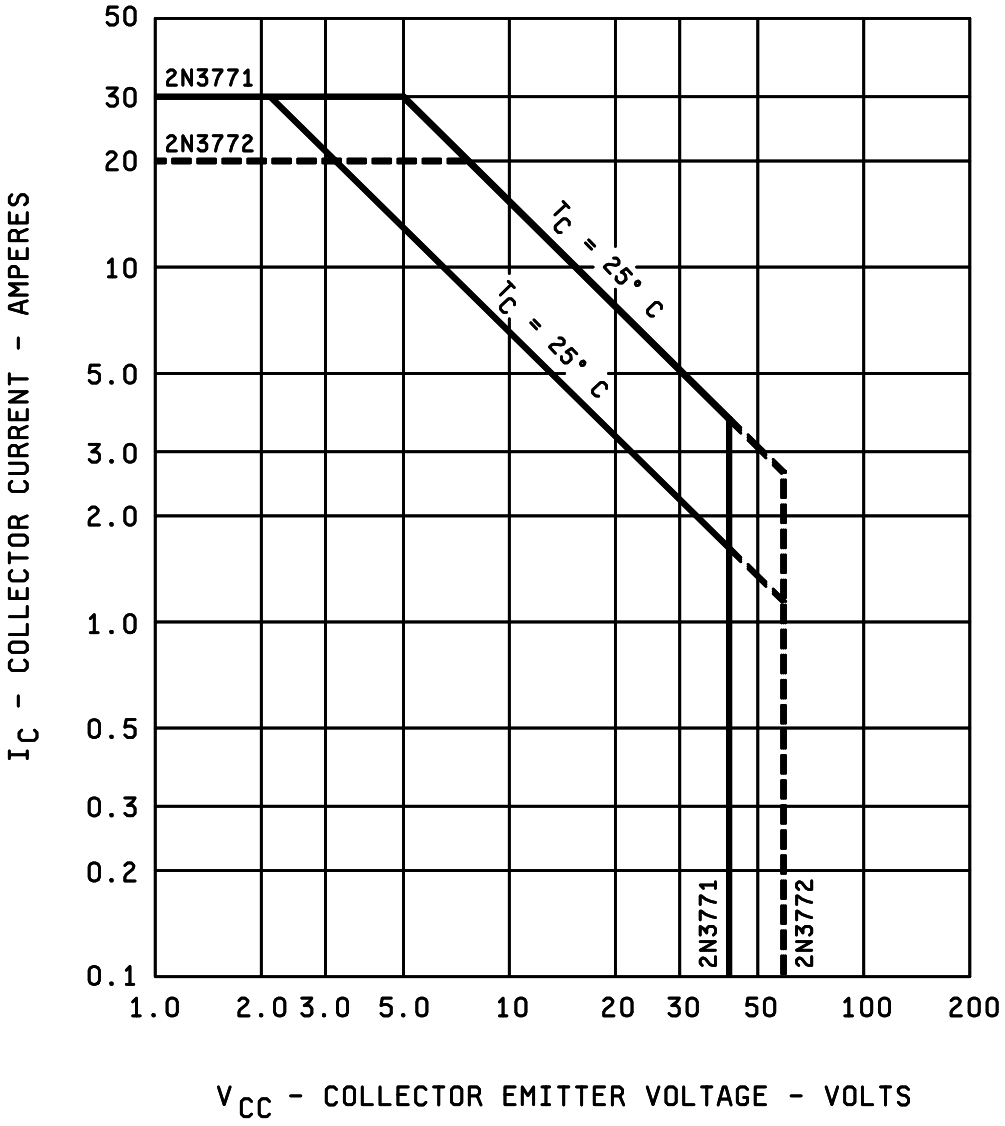


FIGURE 3. Maximum safe operating area graph (continuous dc).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.

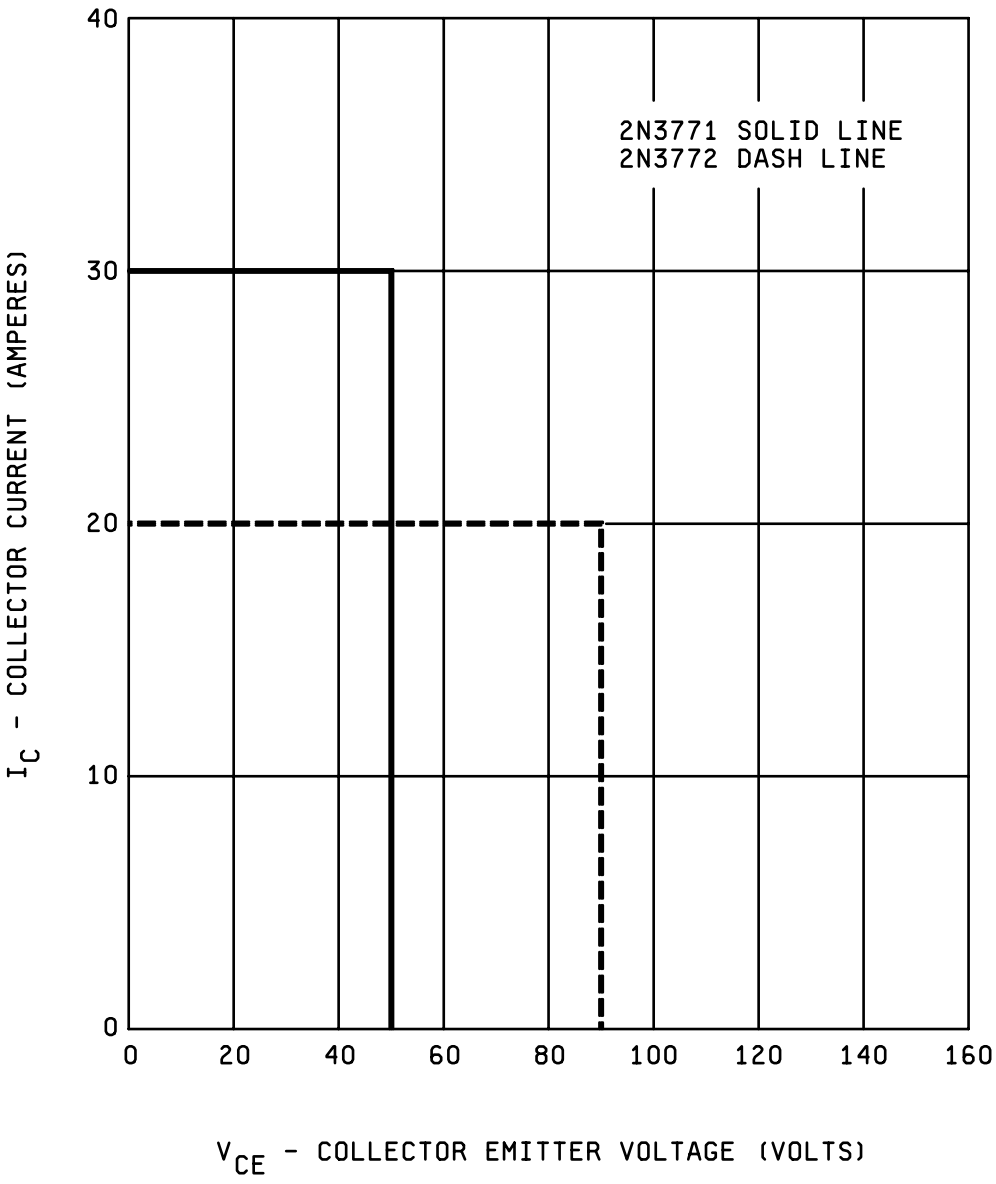


FIGURE 4. Safe operating area for switching between saturation and cutoff (clamped inductive load).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.

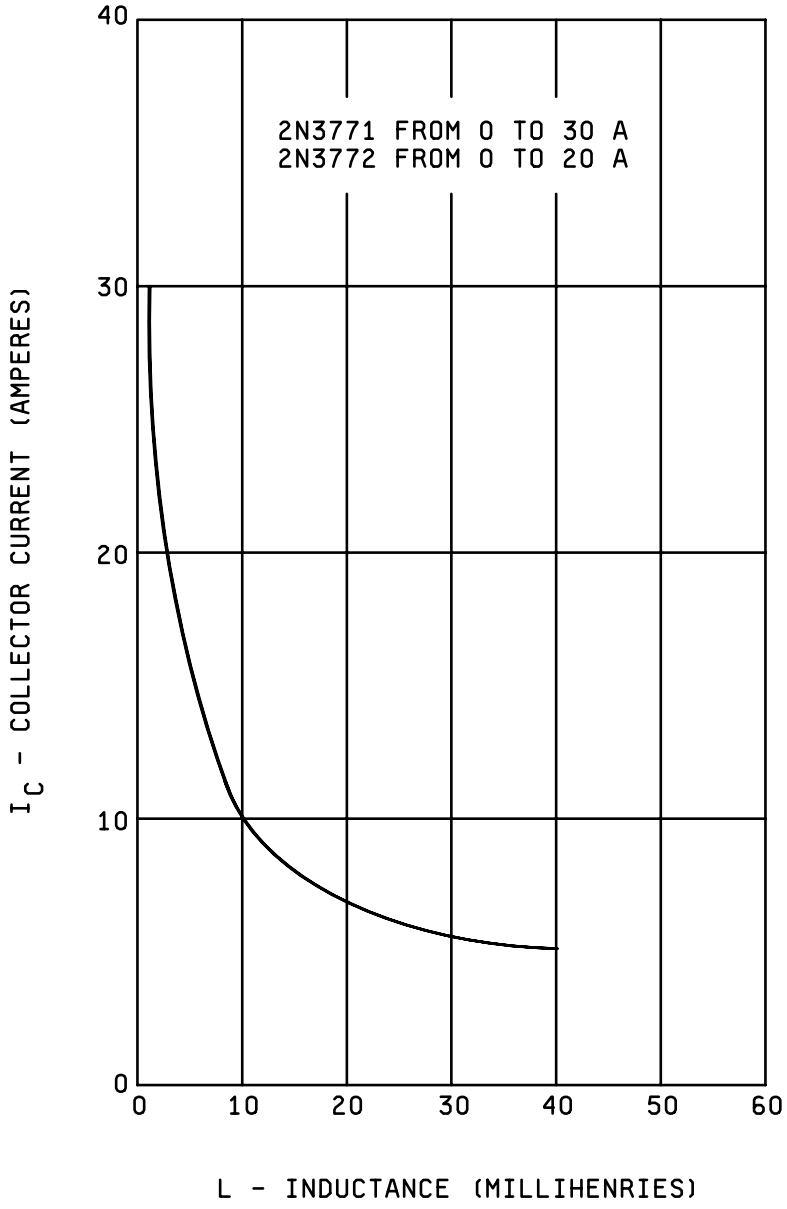


FIGURE 5. Safe operating area switching between saturation and cutoff (unclamped inductive load).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activity:
Air Force - 19, 99
Army-AR

Preparing activity:
DLA - CC

(Project 5961-2007-019)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/ .

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:44Z
Check the source to verify that this is the current version before use.